

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

650 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

650 KWP GRID-TIED SOLAR PLANT

Net investment ₹2.35 Cr after input tax credit, generating 9,96,450 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹4.75 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	650 kWp
Year 1 generation	9,96,450 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹71.35 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹2.47 Cr
GST at 5% — recovered as ITC	₹11.76 L
Net capitalised cost	₹2.35 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹59.20 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,35,23,810	94,09,524	1,41,14,286	23,68,189
2	1,41,14,285	56,45,714	84,68,571	14,20,913
3	84,68,572	33,87,429	50,81,143	8,52,548
4	50,81,143	20,32,457	30,48,686	5,11,529
5	30,48,685	12,19,474	18,29,211	3,06,917
6	18,29,212	7,31,685	10,97,527	1,84,150
7	10,97,527	4,39,011	6,58,516	1,10,490
8	6,58,516	2,63,406	3,95,110	66,294
9	3,95,110	1,58,044	2,37,066	39,777
10	2,37,065	94,826	1,42,239	23,866
11	1,42,240	56,896	85,344	14,320
12	85,343	34,137	51,206	8,592
Tax shield, years 1-12				59,07,585

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

650 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	9,96,450	71,34,582	—	94,09,524	(5,72,557)	77,07,139	77,07,139
2	9,91,468	73,82,865	3,90,000	56,45,714	3,39,051	66,53,814	1,43,60,953
3	9,86,510	76,39,789	3,90,000	33,87,429	9,72,079	62,77,710	2,06,38,663
4	9,81,578	79,05,654	3,90,000	20,32,457	13,80,011	61,35,643	2,67,74,306
5	9,76,670	81,80,771	3,90,000	12,19,474	16,53,864	61,36,907	3,29,11,213
6	9,71,787	84,65,461	3,90,000	7,31,685	18,48,282	62,27,180	3,91,38,393
7	9,66,928	87,60,059	3,90,000	4,39,011	19,96,086	63,73,973	4,55,12,366
8	9,62,093	90,64,910	3,90,000	2,63,406	21,17,007	65,57,902	5,20,70,268
9	9,57,283	93,80,368	3,90,000	1,58,044	22,22,919	67,67,449	5,88,37,717
10	9,52,496	97,06,805	3,90,000	94,826	23,20,988	69,95,818	6,58,33,535
11	9,47,734	1,00,44,602	3,90,000	56,896	24,15,551	72,39,051	7,30,72,586
12	9,42,995	1,03,94,154	3,90,000	34,137	25,09,254	74,94,900	8,05,67,486
13	9,38,280	1,07,55,871	3,90,000	20,482	26,03,727	77,62,143	8,83,29,629
14	9,33,589	1,11,30,175	3,90,000	12,289	26,99,994	80,40,181	9,63,69,810
15	9,28,921	1,15,17,505	3,90,000	7,374	27,98,715	83,28,790	10,46,98,600
16	9,24,276	1,19,18,314	3,90,000	4,424	29,00,333	86,27,982	11,33,26,582
17	9,19,655	1,23,33,072	3,90,000	2,655	30,05,164	89,37,907	12,22,64,489
18	9,15,056	1,27,62,263	3,90,000	1,593	31,13,450	92,58,812	13,15,23,301
19	9,10,481	1,32,06,389	3,90,000	956	32,25,388	95,91,001	14,11,14,302
20	9,05,929	1,36,65,972	3,90,000	573	33,41,152	99,34,819	15,10,49,121
21	9,01,399	1,41,41,547	3,90,000	344	34,60,903	1,02,90,645	16,13,39,766
22	8,96,892	1,46,33,673	3,90,000	206	35,84,796	1,06,58,878	17,19,98,644
23	8,92,408	1,51,42,925	3,90,000	124	37,12,985	1,10,39,940	18,30,38,584
24	8,87,946	1,56,69,899	3,90,000	74	38,45,626	1,14,34,273	19,44,72,857
25	8,83,506	1,62,15,211	3,90,000	45	39,82,878	1,18,42,333	20,63,15,190
26	8,79,088	1,67,79,501	3,90,000	27	41,24,903	1,22,64,598	21,85,79,788
27	8,74,693	1,73,63,427	3,90,000	16	42,71,868	1,27,01,559	23,12,81,347
28	8,70,319	1,79,67,675	3,90,000	10	44,23,947	1,31,53,728	24,44,35,075
29	8,65,968	1,85,92,950	3,90,000	6	45,81,317	1,36,21,633	25,80,56,708
30	8,61,638	1,92,39,984	3,90,000	3	47,44,163	1,41,05,821	27,21,62,529
Total	2,78,24,036	36,70,96,373	1,13,10,000	2,35,23,804	8,36,23,844	27,21,62,529	27,21,62,529

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹4.75 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹27.22 Cr
30-year generation	2,78,24,034 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,78,24,034 units, avoiding 22,815.7 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.47 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 650 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.